



A Holistic Approach to Sustainable, Digital EU Agriculture, Forestry, Livestock and Rural Development based on Reconfigurable Aerial Enablers and Edge Artificial Intelligence-on-Demand Systems

CHAMELEON Bundle User Manual. Crop growth and monitoring. Option 1. Use of multispectral information

Work package	WP4: CHAMELEON Solution for multi-purpose onboarding.
Bundle	Crop growth and monitoring. Option 1. Use of multispectral information
Authors	Miguel A. Moreno – UCLM, David Hernández – UCLM, Rocío Ballesteros – UCLM
Dissemination level	Public (PU)
Status	Draft
Due Date	17/07/2024
Document date	03/07/2024
Version number	0.1
 Funded by the European Union	Funded by the European Union. Views and opinions expressed are however those of the author(s) only and do not necessarily reflect those of the European Union or the European Research Executive Agency. Neither the European Union nor the European Research Executive Agency can be held responsible for them.

Revision and history chart

Version	Date	Main author	Summary of changes
0.1	29/05/2024		Draft outline
0.2	DD/MM/YYYY		
0.3	DD/MM/YYYY		
0.4	DD/MM/YYYY		
0.5	DD/MM/YYYY		
0.6	DD/MM/YYYY		
0.7	DD/MM/YYYY		
0.8	DD/MM/YYYY		
0.9	DD/MM/YYYY		
0.10	DD/MM/YYYY		
0.11	DD/MM/YYYY		
0.12	DD/MM/YYYY		
Etc.	DD/MM/YYYY		(Delete unused lines)
1.0	DD/MM/YYYY		Final submitted version
1.1	DD/MM/YYYY		Revisions after submission (if required)
1.2	DD/MM/YYYY		
2.0	DD/MM/YYYY		Revised submitted version (if required)

Table of contents, select style H1 no ToC

Glossary of terms, Section_Header	4
List of abbreviations and acronyms, Section_Header	4
1 Executive Summary – Select styling H1 TOC.....	5
Exec Sum.....	5
1.1 Purpose and scope.....	5
1.1.1 Crops monitoring using NDVI from multispectral orthomosaic.....	5
1.2 Approach.....	6
1.2.1 uSER INTERFACE OF PAFyc TOOLS AND MAIN PARAMETERS.....	6

Index of figures, select style H1 no ToC

Figure 1: Figure caption, centred, below the figure ¡Error! Marcador no definido.

Index of tables, select style H1 no ToC

Table 1: Table caption, left justified, above the table ¡Error! Marcador no definido.

GLOSSARY OF TERMS

Term	Description

LIST OF ABBREVIATIONS AND ACRONYMS, SECTION_HEADER

Abbreviation	Meaning

1 EXECUTIVE SUMMARY

EXEC SUM

This document outlines a method for monitoring crop growth and development using the Normalized Difference Vegetation Index (NDVI) derived from multispectral orthomosaic images. The primary objective is calculation of NDVI of vine trees by focusing only on vegetation and excluding shadows and soil. The process begins by loading a shapefile containing the frames of the vine trees (referred as "FramesShapefile"), followed by selecting a multispectral orthomosaic file. The method discriminates between soil and vegetation based on an NDVI threshold (minimum NDVI of 0.5) and segments shadows using an RGB threshold (less than 8% in all three visible bands). These NDVI values are then analysed to assess the relative development of vegetation to detect any potential issues such as diseases. To achieve this objective, users can choose between two methods for the vine tree classification: the percentile method and the k-means method. For the percentile method, users must specify band order (i.e. correspondence of each band with the green, blue, red, red-edge and NIR bands in a multispectral camera). After processing, the output shapefile can be loaded into QGIS or WEBGIS app for visualization. The resulting data is categorized by plant health: green for healthy plants (value 0), red for plants with pathology (value 1), and an indication of no plant present (value -1). For the k-means classification method, users set the k-means value up to 6 (depending on the severity searched) and adjust the output file name. The processing and visualization steps are similar to those of the percentile method, with categorization in QGIS or WEBGIS VIEWER based on the generated field values.

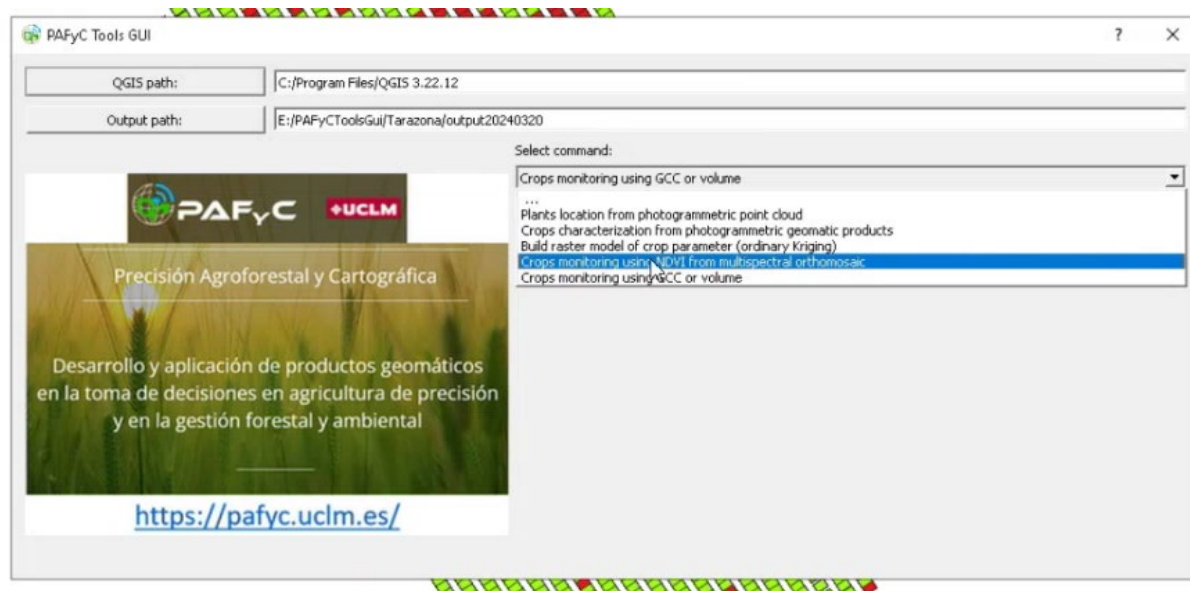
1.1 PURPOSE AND SCOPE

1.1.1 CROPS MONITORING USING NDVI FROM MULTISPECTRAL ORTHOMOSAIC

The objective is the characterization of crop development using the Normalized Vegetation Index (NDVI). The process involves utilizing a specialized NDVI method that excludes shadows and soil and focuses only on vegetation to characterize the growth of each vine tree. NDVI measurements are taken for each frame, allowing for the determination of specific NDVI values. These NDVI values are then analysed to assess the relative development of vegetation to detect any potential issues such as diseases, water stress, or nutritional deficiencies.

1.2 APPROACH

1.2.1 USER INTERFACE OF PAFYC TOOLS AND MAIN PARAMETERS



Firstly, the user needs to introduce the starting file, which is the shapefile that has the frames of the vine trees. In this case, the user should go to the seventh line called the "FramesShapefile" file, which does not have any field values.

Next, the user needs to determine the multispectral file that will be used. So, the user should go the tenth line called "MultispectralOrthomosaicFile". Once the file is loaded, the user can notice there is a date within the file name itself.

Then, the user can, also, choose between two calculation methods: the percentile method or the kmeans method, just like before. In this example, the percentile method was chosen.

The user needs to specify the positions of the bands. For Micasense camera, Blue Band position is 1. Since it is alphabetical, we have Green as 2, Red as 3, and near-infrared band as 5. This command will use NDVI and RGB to classify between what is leaf and what is ground. Shadows are detected based on very low values in the RGB bands present in shadows. So, it does not only use NDVI but also RGB. That is why the user needs to specify the blue and green bands, in addition to the red and infrared.

The discrimination between soil and vegetation is based on the NDVI value, specifically a minimum NDVI. Setting a condition about the minimum NDVI, below 0.5, remove the ground or soil from consideration. However, there is not only soil to be removed, but also shadow and shaded vegetation. To segment shadows, threshold value of RGB should be set. It was found that if it is less than 8% in all three bands (0.08 in the parameter 18), it is assumed to be shadow. Therefore, after elimination of soil and shadow only will be left the NDVI value of vegetation.

CHAMELEON Bundle User Manual. Crop growth and monitoring. Option 1. Use of multispectral information

There is a field called "Date", third line, where the user can manually enter a date. However, there is also a parameter called "Date from Multispectral Orthomosaic File", fifth line, and it is assigned and default to true. So, the program can extract the date from the file name automatically. If the date is included in a specific format in the file name, users should specify a tag in the name (in this case "_") and determine the position of the date information. For example, for "name1_word_20230804_word2.tif", the tag is "_" and the date is in the third position.

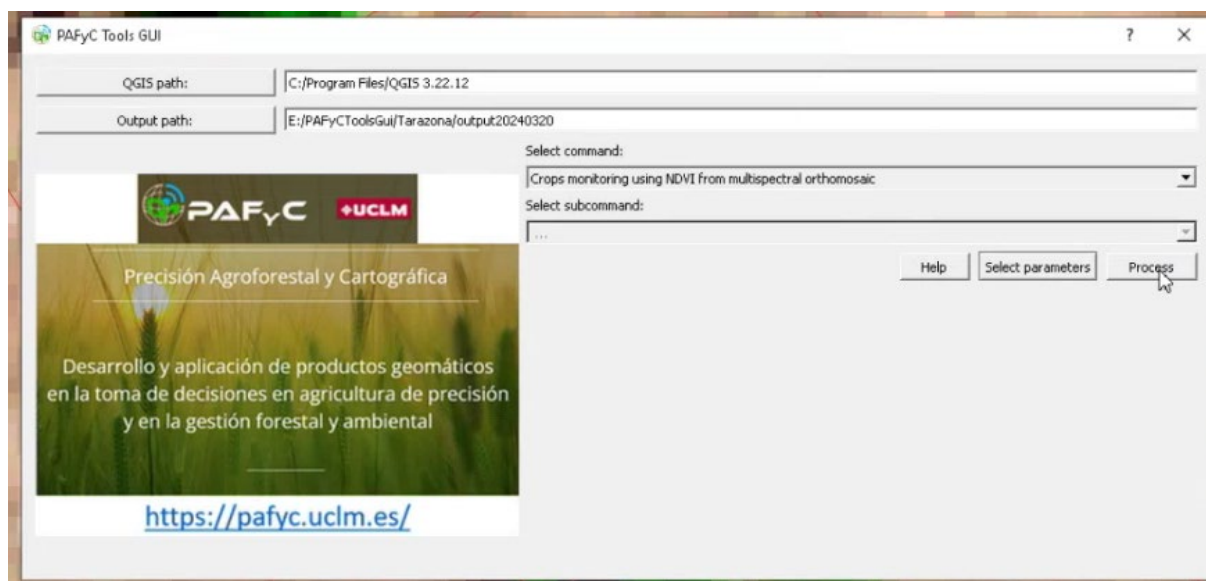
If the percentile method is selected, the user needs to specify the percentile value. For example, if user selects 0.05, the tools will detect as problematic the 5% of vines with lower NDVI values. This value depends on the severity that final user is searching for.

The user needs to specify the path and the output file name, by selecting the output path in the fifteenth line "OutputShapefile".

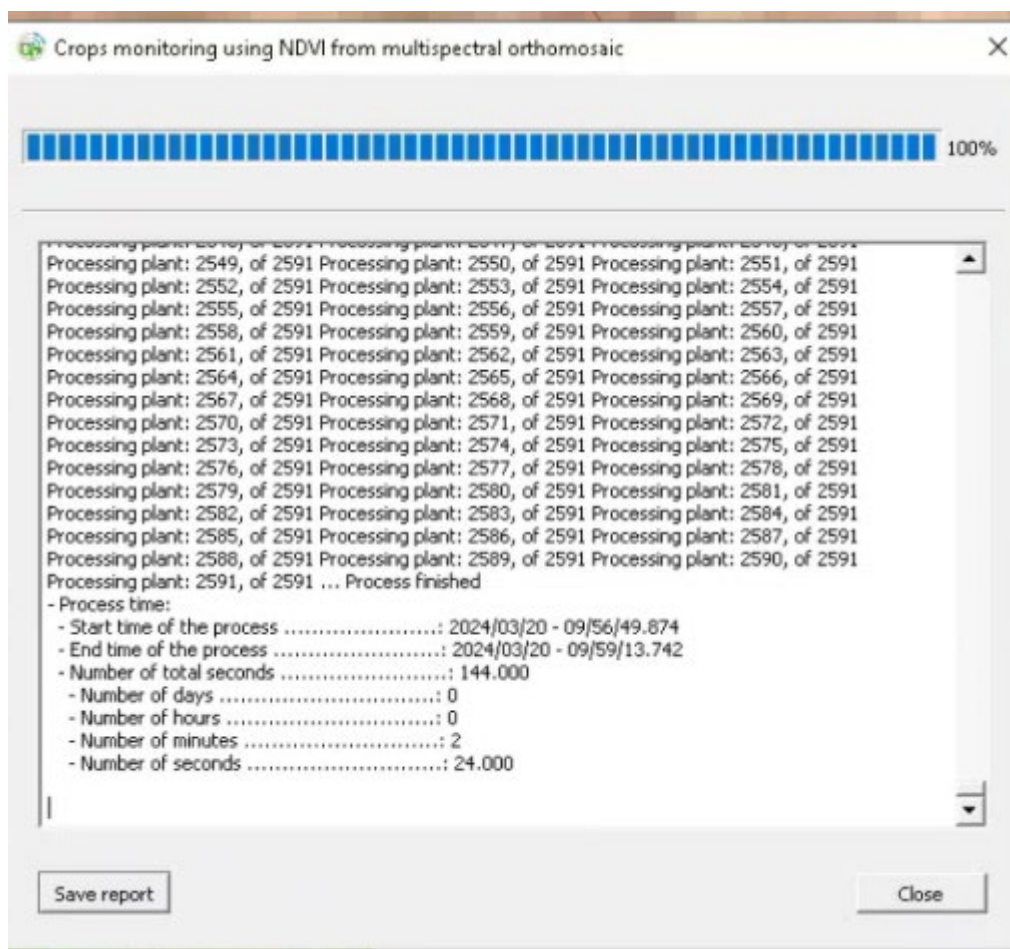
	Code	Enabled	Value	
1	CMNDVI_BlueBandPosition	Yes	1	
2	CMNDVI_CropsMinimumNDVI	Yes	0.50	
3	CMNDVI_Date	Yes		A valid date
4	CMNDVI_DateFormat	Yes	%%Y%%m%%d	
5	CMNDVI_DateFromMultispectralOrthomosaicFile	Yes	True	
6	CMNDVI_FactorToReflectance	Yes	1.000000	Multiplica
7	CMNDVI_FramesShapeFile	Yes	E:/PAFyCToolsGui/Tarazona/20220426_Tarazona_Vid_A6000/input/vines_frames_initial.shp	
8	CMNDVI_GreenBandPosition	Yes	2	
9	CMNDVI_MinimumValueForPercentile	Yes	0.05	
10	CMNDVI_MultispectralOrthomosaicFile	Yes	E:/PAFyCToolsGui/Tarazona/20230630_Tarazona_MULTI_Vid/ORT-303-RFL-Tarazona_Vid_20230630_25830_4cm.tif	
11	CMNDVI_MultispectralOrthomosaicFileTagDatePosition	Yes	3	
12	CMNDVI_MultispectralOrthomosaicFileTagStringSeparator	Yes	-	
13	CMNDVI_NirBandPosition	Yes	5	
14	CMNDVI_NumberOfClustersForKmeans	Yes	6	
15	CMNDVI_OutputShapeFile	Yes	E:/PAFyCToolsGui/Tarazona/20220426_Tarazona_Vid_A6000/output/vines_frames_initial_ndvi_percentile.shp	
16	CMNDVI_RedBandPosition	Yes	3	
17	CMNDVI_SegmentationMethod	Yes	percentile	
18	CMNDVI_SoilRGBMaximumReflectance	Yes	0.08	

Finally, the user should click on process.

CHAMELEON Bundle User Manual. Crop growth and monitoring. Option 1. Use of multispectral information

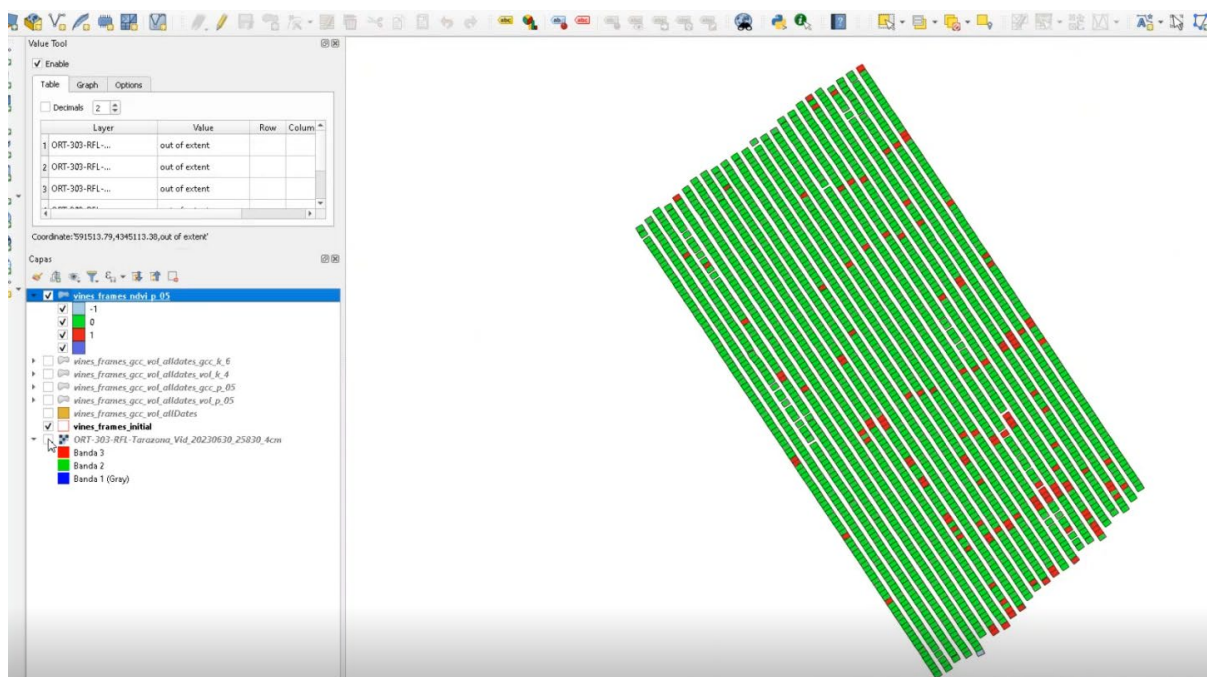


Once the process is completed, the user can go to output path and check the output shapefile.



CHAMELEON Bundle User Manual. Crop growth and monitoring. Option 1. Use of multispectral information

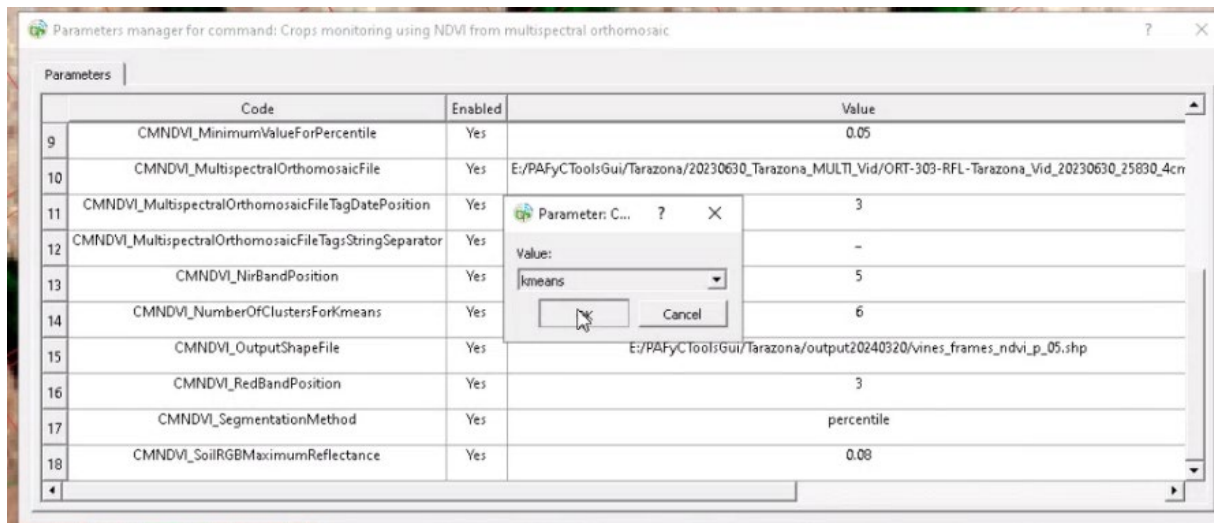
After that, the user can load the result file to QGIS or a WEBGIS viewer. The new shapefile that has been generated and the user can do the same as before: select categorized symbol and select the output generated field. The same as before, the categorization was based on the value of the obtained field. The classification shows that the plants with 0 as those that are healthy, so green colour can be chosen for them, and the plants with a value of 1 would technically have the pathology, so the red colour can be chosen for them. -1 would mean where it appears signifies that there is no plant.



Second example: if we want to choose k-means as classification method.

In this command, also, there is the possibility to choose k-means as classification method, it is known to be slower than percentile method. Compared to the previous example, no need to change any parameters other than specifying the selection of K-means method with a value of 6 or lower (depending on the severity searched) and keeping all other settings as they are in the previous example. Additionally, the user needs to update the name of the output file for the results.

CHAMELEON Bundle User Manual. Crop growth and monitoring. Option 1. Use of multispectral information

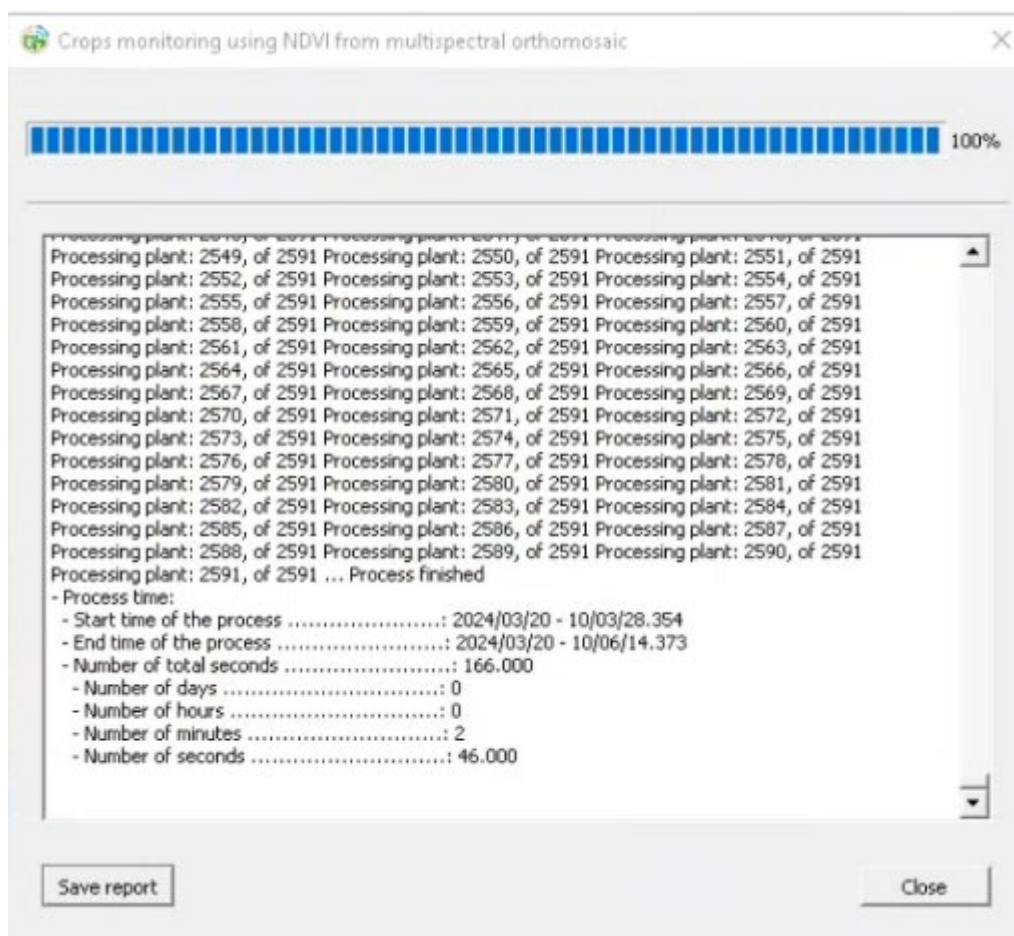


Finally, the user should click on process.



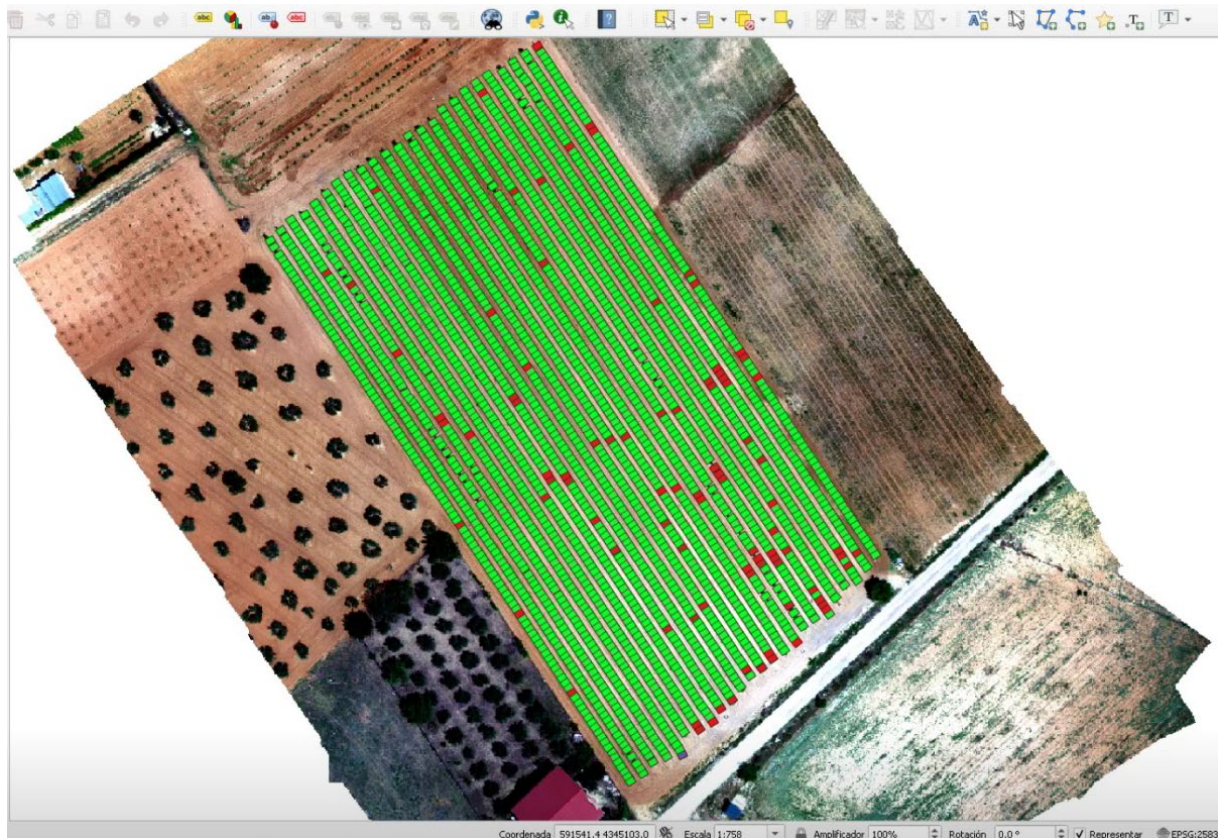
Once the process is completed, the user can go to output path and check the output shapefile.

CHAMELEON Bundle User Manual. Crop growth and monitoring. Option 1. Use of multispectral information



After that, the user can load the result file to QGIS or a WEBGIS. The new shapefile that has been generated and the user can do the same as before: select categorized symbol and select the output generated field. The same as before, the categorization was based on the value of the obtained field. The classification shows that the plants with 0 as those that are healthy, so green colour can be chosen for them, and the plants with a value of 1 would technically have the pathology, so the red colour can be chosen for them. -1 would mean where it appears signifies that there is no plant.

CHAMELEON Bundle User Manual. Crop growth and monitoring. Option 1. Use of multispectral information



Copyright © 2022. All rights reserved.



A Holistic Approach to Sustainable, Digital EU Agriculture, Forestry, Livestock and Rural Development based on Reconfigurable Aerial Enablers and Edge Artificial Intelligence-on-Demand Systems

The Members of the CHAMELEON Consortium:



Contact:



Project Coordinator: Pantelis Velanas Accelligence Ltd.	pvelanas@accelligence.tech
---	--

Disclaimer

Funded by the European Union. Views and opinions expressed are however those of the author(s) only and do not necessarily reflect those of the European Union or the European Research Executive Agency. Neither the European Union nor the European Research Executive Agency can be held responsible for them.